

YI ZHANG

PhD in Applied Computer Science

Augsburg | 015236410279 | zhangyi1353173@yahoo.com
https://www.linkedin.com/in/yi-zhang-2020/

Soon-to-graduate PhD specializing in probabilistic deep learning for real-world industrial applications. Experienced in bridging state-of-the-art AI research with practical deployment, with a focus on developing reliable, robust, and efficient system identification and state estimation methods – essential for system observation, control, and decision making

Education

University of Augsburg 04.2021-03.2026 (estimated)
Doctor of Engineering in Applied Computer Science
Title of Dissertation: Probabilistic System Identification and State Estimation for Dynamical Systems via Normalizing Flows
https://www.uni-augsburg.de/de/fakultaet/fai/informatik/prof/imech/team/yi-zhang/

University of Stuttgart 10.2017 – 09.2020
Master of Science in Electrical and Electronics Engineering
Major: Signal processing, Machine Learning
Master Thesis: Feature Selection using Attention Mechanism in Deep Learning
Grade: 1.6

TH Aschaffenburg University of Applied Science 09.2016 – 09.2017
Bachelor of Engineering in Mechatronics (Exchange Program)
Major: Drive technology and robotics, Application of micro-electrotechnology
Bachelor Thesis: Algorithms for avoiding singularity in industrial robots
Grade: 1.5

University of Tongji, Shanghai 09.2013 – 09.2016
Bachelor of Engineering in Mechatronics
Grade: 1.8

Experience

Trumpf Werkzeugmaschinen GmbH, Ditzingen 05.2019 – 10.2019
Intern in R&D Pre-development Production Platform MT
- Support in the development of a prediction model for process-reliable part removal on a fully automated laser cutting machine
- Further development and implementation of a novel computer vision approach for teaching a finite element structure simulation through U-Net architectures, with Keras
- Evaluation and training of the models as well as realization and identification of improvement potentials to increase model accuracy

Publications

Zhang, Y. and Mikelsons, L., IMFlow: Inverse Modeling with Conditional Normalizing Flows for Data-Driven Model Predictive Control, 2024 IEEE 63rd Conference on Decision and Control (CDC), Milan, Italy, 2024, pp. 3353-3359, doi: 10.1109/CDC56724.2024.10886862.
Zhang, Y. and Mikelsons, L., Solving Stochastic Inverse Problems with Stochastic BayesFlow, 2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Seattle, WA, USA, 2023, pp. 966-972, doi: 10.1109/AIM46323.2023.10196190.
Zhang, Y. and Mikelsons, L., Sensitivity-guided iterative parameter identification and data generation with BayesFlow and PELS-VAE for model calibration. Adv. Model. and Simul. in Eng. Sci. 10, 9 (2023). doi: 10.1186/s40323-023-00246-y.
Heringhaus, M.E.; Zhang, Y.; Zimmermann, A.; Mikelsons, L., Towards Reliable Parameter Extraction in MEMS Final Module Testing Using Bayesian Inference. Sensors 2022, 22, 5408. doi: 10.3390/s22145408

Programming Languages

Python, Julia
MATLAB/Simulink, C++

Languages

English: Professional Proficiency
German: Professional Proficiency
Chinese: Native Proficiency

Technical Skills

Machine Learning/Deep Learning
Reinforcement Learning
Mathematics & Statistics
Data processing & -analysis
ML-Frameworks: PyTorch, Jax, TensorFlow/Keras

Problem-solving expertise

Implementation and optimization of algorithms
Interdisciplinary work
Creativity & critical thinking
Ability to think abstractly

Soft Skills

Explanation of complex concepts
Collaboration with researchers and industry partners
Independence & personal responsibility
Time management & prioritisation
Teamwork & collaboration
Openness to feedback
Curiosity & lifelong learning
Perseverance

Third-party funded projects

PHyMoS 04.2021 – 02.2024
(https://phymos.de/)
OpenScaling 11.2023 – 10.2026
(https://openscaling.org/)